

4th Chapter

Islam Modern Science

Quran and Modern Science
Contribution of Muslim Scientists

Ms. SAFIA WAQAS

Session: Spring: 2025

Course: Islamic Studies / Ethical Behavior



Quran and Modern Science

► Topics:

- Introduction Quran and Modern Science
- Scientific Discoveries That Match Quranic Teachings
 - A. The Development of the Human Embryo
 - B. The Development of the Human Embryo
 - C. The Origin of the Universe
 - D. The Role of Water in Life
 - E. The Protective Layers of the Atmosphere
 - F. Mountains as Pegs
 - G. The Barrier Between Salt and Fresh Water
 - H. The Stages of Clouds and Rain Formation
- Quranic Insights: Evidence of Divine Origin and Scientific Accuracy

Introduction Quran and Modern Science

- Modern science is now finding facts and discoveries that match the descriptions given in the Holy Quran.
- This is very amazing because the Quran was revealed over 1,400 years ago, long before modern science existed.
- Many people believe these scientific discoveries are strong evidence that the Quran is a divine revelation from Allah.

Main Question:

How is modern science proving the authenticity of the Holy Quran?

Scientific Discoveries That Match Quranic Teachings

A. The Development of the Human Embryo

Quranic Description:

The Holy Quran explains the growth of a human in clear stages. It starts with a tiny sperm-drop, then becomes a clinging clot, next a small lump, and finally forms bones that are covered by flesh.

“We created man from an extract of clay. Then We placed him as a sperm-drop in a firm lodging. Then We made the sperm-drop into a clinging clot, and We made the clot into a lump [of flesh], and We made [from] the lump, bones, and We covered the bones with flesh; then We developed him into another creation. So blessed is Allah, the best of creators.”

(Surah Al-Mu'minun, 23:12–14)

Scientific Explanation:

Modern science, using microscopes and imaging tools, has confirmed these stages of growth. Experts like Dr. Keith L. Moore have observed that the steps described in the Quran match the real process of human development very closely.

What Science Finds:

Today, scientists can see and study these changes using special tools like microscopes. They have found that humans do indeed grow in these exact steps, which is very surprising because this was known over 1,400 years ago.

Why It Matters:

This match between the Quran and modern science is important because it shows that the Quran contains knowledge that was ahead of its time, reinforcing the belief that it is a divine message from Allah.

B. The Development of the Human Embryo

Quranic Description:

The Quran tells us that the universe is growing bigger. It says:

“And the heaven We constructed with strength, and indeed, We are [its] expander.”
(Surah Adh-Dhariyat, 51:47)

Scientific Explanation:

In the 1920s, scientist Edwin Hubble discovered that galaxies are moving away from each other. This means the universe is expanding—just as the Quran describes.

What the Quran Says:

The Quran tells us that the universe was made strong and is still growing.

What Science Found:

Edwin Hubble’s work showed that galaxies are moving apart, proving that the universe is indeed expanding.

Why It Is Important:

This matching of the Quranic message with modern scientific discovery is seen by many as evidence that the Quran is a divine message from Allah.

C. The Origin of the Universe

Quranic Description:

The Quran tells us that the heavens (the sky and everything beyond) and the earth were once a single, connected whole. Then, Allah separated them. The verse says:

“Have those who disbelieved not considered that the heavens and the earth were a joined entity, and We separated them and made from water every living thing? Then will they not believe?”

(Surah Al-Anbiya, 21:30)

Scientific Explanation:

Modern science explains the beginning of the universe with the Big Bang theory. According to this theory, the universe started as a very small, dense point called a singularity. It then expanded rapidly and eventually split into stars, planets, galaxies, and everything else we see today.

What the Quran Says:

The Quran explains that in the past, the heavens and the earth were together as one. Allah then separated them, which is a key sign for people to believe in His power.

What Science Found:

Scientists believe that the universe began with a Big Bang, where all matter was once squeezed into a tiny point. This point expanded and broke apart, forming all the celestial bodies like stars and planets.

Why It Matters:

The similarity between the Quranic description and the Big Bang theory suggests that the Quran contains knowledge that aligns with modern scientific discoveries. This is seen by many as evidence of the divine origin of the Quran.

D. The Role of Water in Life

Quranic Description:

The Quran highlights that water is the source of all life.

“And We made from water every living thing. Then will they not believe?”
(Surah Al-Anbiya, 21:30)

Scientific Explanation:

Modern biology shows that water is very important for life. Every living thing needs water for the chemical reactions that keep it alive, which confirms what the Quran says.



E. The Protective Layers of the Atmosphere

Quranic Description:

The Quran describes the sky as a protective ceiling that shields the Earth.

“And We made the sky a protected ceiling, but they, from its signs, are turning away.”

(Surah Al-Anbiya, 21:32)

Scientific Explanation:

Modern science shows that the Earth's atmosphere acts like a protective shield. It blocks harmful rays from the sun, helps keep our temperature steady, and even protects us from meteors hitting the planet—just as the Quran describes.

F. Mountains as Pegs

Quranic Description:

The Quran compares mountains to pegs or stakes that help keep the Earth steady. It says:
“Have We not made the earth a resting place? And the mountains as stakes?”
(Surah An-Naba, 78:6–7)

Scientific Explanation:

Modern geology shows that mountains have deep roots in the Earth, much like pegs. These roots help to hold the Earth's surface stable, which matches the Quran's description.

G. The Barrier Between Salt and Fresh Water

Quranic Description:

The Quran mentions that when two seas meet, a barrier exists between them so that they do not mix completely.

“He released the two seas, meeting [side by side]; Between them is a barrier [so] neither of them transgresses.”

(Surah Ar-Rahman, 55:19–20)

Scientific Explanation:

Oceanographers have found that when saltwater and freshwater meet, they do not mix completely because they have different amounts of salt and weight. This creates a clear boundary called a halocline, which matches the Quranic description.

H. The Stages of Clouds and Rain Formation

Quranic Description:

The Quran explains how winds and clouds work together to bring rain. It says:

“And it is He who sends the winds as good tidings before His mercy, and We send down from the sky pure water.”

(Surah Al-Furqan, 25:48)

Scientific Explanation:

Meteorologists tell us that winds carry water vapor, which gathers to form clouds. When these clouds become heavy, they release rain. This natural process is described very accurately in the Quran.

Quranic Insights: Evidence of Divine Origin and Scientific Accuracy

A. Evidence of Divine Origin

The Quran gives very clear and accurate descriptions of nature. Even though it was revealed more than 1,400 years ago, these details match what modern science has discovered. This strongly shows that the Quran is from Allah.

B. Encouragement of Inquiry

The Quran invites us to look at the world around us. It asks us to observe and think about nature and its amazing details. This helps us learn more about the wonders of creation.

“Do they not reflect upon the creation of the heavens and the earth?” [\(Surah Al-Imran, 3:190\)](#)

C. Integration of Faith and Knowledge

The matching facts between the Quran and modern science inspire Muslims to explore and learn more about the world. This connection shows that our faith and scientific study can work together, helping us to grow in both knowledge and spirituality.

Modern scientific discoveries confirm many descriptions found in the Quran. Whether it is the stages of human development, the expanding universe, or the protective role of the atmosphere, these facts support the divine origin of the Quran.

While the Quran is not a science textbook, its clear and accurate references to natural phenomena show its timeless relevance. They inspire both spiritual reflection and a quest for knowledge, encouraging us to see the hand of Allah in every part of the universe.

Islam Modern Science – Contribution of Muslim Scientists

► Topics:

- Introduction
- Islam's Encouragement of Knowledge
 - A. Quranic Inspiration
 - B. Hadiths on Knowledge
- Major Contributions by Muslim Scientists
 - A. Astronomy
 - B. Medicine
 - C. Mathematics
 - D. Physics and Engineering
 - E. Chemistry
 - F. Geography
- Integration of Quranic Teachings with Science
- Scientific Legacy of Muslim Scholars

Islam Modern Science - Contribution of Muslim Scientists

Muslim scientists played a major role in advancing science by blending the wisdom of the Quran with their strong desire to learn about the world. Inspired by the Quran and the teachings of Prophet Muhammad (peace be upon him), they explored many fields such as astronomy, medicine, mathematics, physics, chemistry, and geography during the Islamic Golden Age (8th to 14th century CE). Their discoveries and inventions laid the foundation for modern science and continue to benefit us today.

Islam's Encouragement of Knowledge

➤ Quranic Inspiration

The Quran encourages us to observe the natural world and think about its amazing details.

“Do they not look into the realm of the heavens and the earth and all that Allah has created?” (Surah Al-A'raf, 7:185)

“Read in the name of your Lord who created...” (Surah Al-Alaq, 96:1)

These verses remind us that seeking knowledge and understanding the world is very important.

➤ Hadiths on Knowledge

The Prophet Muhammad (peace be upon him) highlighted the value of learning:

“Seeking knowledge is an obligation upon every Muslim.” (Ibn Majah)

“Wisdom is the lost property of the believer; wherever they find it, they are most deserving of it.” (Sunan at-Tirmidhi)

Major Contributions by Muslim Scientists

Astronomy

Quranic Inspiration:

The Quran invites us to look at the sky and learn about its wonders. It tell us

“And it is He who created the night and the day and the sun and the moon; all [heavenly bodies] in an orbit are swimming.”

(Surah Al-Anbiya, 21:33)

Contributions:

Al-Battani (858-929 CE):

He measured the length of the solar year with great accuracy.

His work later influenced European astronomers like Copernicus.

Al-Zarqali (1029-1087 CE):

He designed advanced astrolabes, tools used to study the stars.

He improved older models of the universe, helping to shape new ideas about how celestial bodies move.

Medicine

Contributions:

Al-Razi (Rhazes) (865-925 CE):

Wrote Kitab al-Hawi (Comprehensive Book of Medicine), a detailed medical encyclopedia used in Europe for many years.

He was the first to clearly tell the difference between smallpox and measles.

Ibn Sina (Avicenna) (980-1037 CE):

Wrote Al-Qanun fi al-Tibb (The Canon of Medicine), a very important book in medicine and pharmacy.

He introduced new ideas, such as quarantine, to help stop the spread of diseases.

Al-Zahrawi (Albucasis) (936-1013 CE):

Developed many new surgical tools and techniques.

His 30-volume work, Kitab al-Tasrif, became a key reference for surgery.



Major Contributions by Muslim Scientists

Mathematics

Contributions:

Al-Khwarizmi (780-850 CE):

Known as the "father of algebra." His book introduced algebra to the world and gave us the word "algorithm."

Omar Khayyam (1048-1131 CE):

He developed ways to solve cubic equations and helped create a more accurate solar calendar.

Thabit ibn Qurra (826-901 CE):

He made key advances in geometry, trigonometry, and number theory.

Physics and Engineering

Contributions:

Ibn al-Haytham (Alhazen) (965-1040 CE):

He wrote Kitab al-Hawi (Comprehensive Book of Medicine), a complete medical encyclopedia that was used in Europe for many years.

He also was the first to clearly show the difference between smallpox and measles. **Ibn Sina (Avicenna) (980-1037 CE):**

He wrote Al-Qanun fi al-Tibb (The Canon of Medicine), an important book on medicine and drugs.

He introduced new ideas such as quarantine to help stop the spread of diseases.

Al-Zahrawi (Albucasis) (936-1013 CE):

He developed many new surgical tools and techniques. His 30-volume encyclopedia, Kitab al-Tasrif, became a key guide for surgeons and helped shape modern surgery.

Major Contributions by Muslim Scientists

Chemistry

Contributions:

Jabir ibn Hayyan (Geber) (721-815 CE):

Known as the "father of chemistry," he was the first to use systematic experiments to study chemicals.

He invented important techniques such as distillation, crystallization, and evaporation.

He wrote extensively about acids and various chemical substances, laying the groundwork for modern chemistry.

Geography

Contributions:

Al-Idrisi (1100-1165 CE):

He created very detailed maps and a world atlas for King Roger II of Sicily. His work combined knowledge from both Islamic and European sources.

Ibn Battuta (1304-1369 CE):

He traveled over 75,000 miles and wrote about the different cultures he encountered in his travel book, Rihla (The Journey).

Integration of Quranic Teachings with Science

A. Observation and Reflection

- The Quran encourages us to look closely at nature.
- This inspired Muslim scientists to study the world around them carefully.

B. Unity of Knowledge

- Muslim scientists believed that faith and science work together.
- They thought that learning about nature helps us understand Allah's creation.

C. Practical Applications

- They used what they learned to make life better.
- Their discoveries improved medicine, helped build stronger structures, and improved how communities were run.
- This work matches the Quran's message of helping and serving people.

How Muslim Scholars Changed the World

A. Impact on the European Renaissance

- Many works by Muslim scholars were translated into Latin and taught in European universities. This helped start the European Renaissance, with thinkers like Roger Bacon and Galileo learning from these contributions.

B. Saving of Ancient/earlier Knowledge

- Muslim scholars preserved and expanded the knowledge of the Greeks, Romans, and Persians during Europe's Dark Ages, ensuring that this wisdom was not lost.

C. Joining Faith and Reason

- The Islamic Golden Age showed that faith and reason can work together. This legacy continues to inspire modern efforts to mix ethical values with scientific research.

The work of Muslim scientists shows the deep impact of Islamic teachings on the advancement of knowledge. Their discoveries in astronomy, medicine, mathematics, physics, chemistry, and geography were not only groundbreaking but also deeply connected with the teachings of the Quran.

Inspired by the Quran and Hadith, these scholars set high ethical and intellectual standards that continue to benefit humanity today. Their legacy demonstrates that faith and science can work together to promote human welfare and help us understand the wonders of creation.